SECR2043 OPERATING SYSTEMS

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Section : 02

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

1.	Display your current working directory.	
	Command	**Output**
	pwd	chew@secr2043:~$ pwd /home/chew
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	**Output**
	ls -a ~	chew@secr2043:~$ ls -a ~ . .local file2.txt practice.txt .. .profile filelist.txt sample.txt .bash_logout .sudo_as_admin_successful hello.txt workspace .bashrc combined.txt linux-lab
3.	Create a new directory named "os_lab" in your home directory.	
	Command	**Output**
	mkdir ~/os_lab	No Output
4.	Change your current working directory to "os_lab".	
	Command	**Output**
	cd ~/os_lab	No Output
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	**Output**
	echo "Hello, world!" > hello.txt	No Output

<table>
<tr><td>6.</td><td colspan="2">Display the content of "hello.txt".</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>cat hello.txt</td><td>chew@secr2043:~/os_lab$ cat hello.txt
Hello, world!</td></tr>
<tr><td>7.</td><td colspan="2">Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>cp hello.txt hello_copy.txt</td><td>No Output</td></tr>
<tr><td>8.</td><td colspan="2">Rename "hello_copy.txt" to "hello_bak.txt".</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>mv hello_copy.txt hello_bak.txt</td><td>No Output</td></tr>
<tr><td>9.</td><td colspan="2">Delete "hello.txt".</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>rm hello.txt</td><td>No Output</td></tr>
<tr><td>10.</td><td colspan="2">Display your home directory with additional information, together with subdirectories contents.</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>ls -lR ~</td><td>chew@secr2043:~/os_lab$ ls -lR ~
/home/chew:
total 36
-rw-r--r-- 1 chew chew 158 Apr 16 13:53 combined.txt
-rw-r--r-- 1 chew chew 79 Apr 16 13:48 file2.txt
-rw-r--r-- 1 chew chew 364 Apr 16 13:57 filelist.txt
-rw-r--r-- 1 chew chew 64 Apr 16 14:04 hello.txt
drwxr-xr-x 2 chew chew 4096 Apr 16 14:41 linux-lab
drwxr-xr-x 2 chew chew 4096 Apr 16 14:52 os_lab
-rw-r--r-- 1 chew chew 718 Apr 16 14:16 practice.txt
-rw-r--r-- 1 chew chew 814 Apr 16 14:14 sample.txt
drwxr-xr-x 2 chew chew 4096 Apr 16 14:31 workspace

/home/chew/linux-lab:
total 0

/home/chew/os_lab:
total 4
-rw-r--r-- 1 chew chew 14 Apr 16 14:52 hello_bak.txt

/home/chew/workspace:
total 44
-rwxr-xr-x 1 chew chew 16008 Apr 16 14:31 arguement
-rw-r--r-- 1 chew chew 398 Apr 16 14:31 arguement.c
-rwxr-xr-x 1 chew chew 15968 Apr 16 14:21 hello_world
-rw-r--r-- 1 chew chew 76 Apr 16 14:18 hello_world.c
-rw-r--r-- 1 chew chew 94 Apr 16 14:23 hello_world.cpp</td></tr>
<tr><td></td><td>Total Mark:</td><td></td></tr>
</table>

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

<table>
<tr><td rowspan="3">1.</td><td colspan="2">Create an empty file named "hello.txt" in the current working directory.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>touch hello.txt</td><td>No Output</td></tr>
<tr><td rowspan="3">2.</td><td colspan="2">Write the text "Hello World" to "hello.txt" using echo.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>echo "Hello World" > hello.txt</td><td>No Output</td></tr>
<tr><td rowspan="3">3.</td><td colspan="2">Append the text "This is a test" to "hello.txt" using echo.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>echo "This is a test" >> hello.txt</td><td>No Output</td></tr>
<tr><td rowspan="3">4.</td><td colspan="2">Display the contents of "hello.txt" using cat.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>cat hello.txt</td><td>chew@secr2043:~/os_lab$ cat hello.txt
Hello World
This is a test</td></tr>
<tr><td rowspan="3">5.</td><td colspan="2">Create another file named "bye.txt" with the text "Goodbye World" using echo.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>echo "Goodbye World" > bye.txt</td><td>No Output</td></tr>
<tr><td rowspan="3">6.</td><td colspan="2">Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>cat hello.txt bye.txt</td><td>chew@secr2043:~/os_lab$ cat hello.txt bye.txt
Hello World
This is a test
Goodbye World</td></tr>
</table>

7. Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.

Command	Output
cat hello.txt bye.txt > greeting.txt	No Output

8. Display the contents of "greeting.txt" using cat.

Command	Output

cat greeting.txt	`chew@secr2043:~/os_lab$ cat greeting.txt`
	`Hello World`
	`This is a test`
	`Goodbye World`

9. Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.

Command	Output

nano greeting.txt
`GNU nano 7.2`
`Hello Chew`
`This is a test`
`Goodbye Chew`
	No Output

10. Display the modified contents of "greeting.txt" using cat.

Command	Output

cat greeting.txt	`chew@secr2043:~/os_lab$ cat greeting.txt`
	`Hello Chew`
	`This is a test`
	`Goodbye Chew`

Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc.
For each question, write down the command and the output (if any) in the answer sheet.

1. Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".

Command	Output
nano hello.c	No Output

```
  GNU nano 7.2
#include <stdio.h>
int main() {
    printf("Hello World\n");
    return 0;
}
```

2. Compile the program "hello.c" using gcc and generate an executable file named "hello".

Command	Output
gcc hello.c -o hello	No Output

3. Run the executable file "hello" and display its output.

Command	Output
./hello	chew@secr2043:~/os_lab$ ./hello Hello World

4. Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".

Command	Output
nano sum.c	No Output

```
 GNU nano 7.2                                sum.c
#include <stdio.h>
#include <stdlib.h>
int main(int argc, char *argv[]) {
    if (argc < 3) {
        printf("Usage: %s num1 num2\n", argv[0]);
        return 1;
    }
    int a = atoi(argv[1]);
    int b = atoi(argv[2]);
    printf("Sum: %d\n", a + b);
    return 0;
}
```

5. Compile the program "sum.c" using gcc and generate an executable file named "sum".

Command	Output
gcc sum.c -o sum	No Output

6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.

Command	Output
./sum 10 20	chew@secr2043:~/os_lab$ ./sum 10 20 Sum: 30

7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.

Command	Output
./sum -5 15	chew@secr2043:~/os_lab$ ./sum -5 15 Sum: 10

8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".

Command	Output
nano echo.c	No Output

```
  GNU nano 7.2
#include <stdio.h>
int main() {
    char str[100];
    printf("Enter text: ");
    fgets(str, sizeof(str), stdin);
    printf("The text entered is: %s", str);
    return 0;
}
```

9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".

Command	Output
gcc echo.c -o echo	No Output

10.	Run the executable file "echo", type "Hello World" as input and display its output.

Command	Output
./echo	chew@secr2043:~/os_lab$ ./echo Enter text: Hello World The text entered is: Hello World

Total Mark: